New challenges for an NSBI measurement of κ_λ in HH$\to b\bar{b}\tau\tau$, relative to the first NSBI measurement (ATLAS off-shell $H \to ZZ \to 4\ell$)

working note — draft for discussion

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Scope & one-line conclusion

This compares the systematics and background handling of the **first full NSBI measurement** (ATLAS off-shell $H \to ZZ \to 4\ell$; note HIGG-2023-01 = arXiv:2412.01600, Sandesara *et al.*) with what a κ_λ **measurement in HH**$\to b\bar{b}\tau\tau$ must do (baselined on CMS AN-18-121, Amendola *et al.*). The NSBI *skeleton* transfers directly; the new difficulty is almost entirely in the *inputs*: $b\bar{b}\tau\tau$ has a far messier final state, several **data-driven backgrounds with no per-event MC density** (QCD, jet$\to\tau$ fakes), an **irreducible same-final-state single-Higgs background**, an order of magnitude more (and larger) **experimental systematics that deform the very observable (m_{HH}) carrying the κ_λ signal**, and a **κ_λ-dependent theory uncertainty**. None of these are NSBI-specific — they hit any method — but several sit awkwardly inside a per-event density-ratio framework in ways the clean, all-MC off-shell analysis never faced. Page references to both notes are given inline.

Studies to do

Concrete studies to de-risk and execute this measurement, grouped by theme. Priority: **[P0]** blocking / do-first, **[P1]** important, **[P2]** nice-to-have; effort (S <1 wk, M weeks, L months). Trailing [§n] point to the relevant numbered section below. All abbreviations are defined in the **Glossary** at the end. Status tags: **[✓ done]** = completed and **[partial]** = partially completed in the preliminary phenomenological study (see the Progress box).

Progress — preliminary phenomenological result (Ghosh, Klute & Pan, 22 Jun 2026)

A first $b\bar{b}\tau\tau$ NSBI study (code in `NSBI-pheno/dihiggs_bbtautau`) has completed several items below. *The method and sensitivity are validated on **CMS full-simulation** samples (POWHEG NLO signal), but **without experimental-systematic nuisances in the fit**; a publishable phenomenological version additionally needs a DELPHES fast-sim reproduction (in progress — item #27, and the note's described setup). So the statistics and physics-method items are largely done, while the systematics, in-situ-CR, data-driven-background and Delphes-sample items remain open.* **Headline:** an unbinned 10-feature NSBI likelihood lifts the binned-m_{HH} secondary minimum near $\kappa_\lambda \sim 3$–4 and **narrows the 68% interval by ∼an order of magnitude**, raising the $\kappa_\lambda = 0$ rejection from 1.1σ (binned m_{HH}) to $\mathbf{2.75\sigma}$ at $450\,\mathrm{fb}^{-1}$ and to $\mathbf{7.1\sigma}$ at the HL-LHC ($3000\,\mathrm{fb}^{-1}$, vs 2.8σ binned); 500-toy pulls $\sim \mathcal{N}(0,1)$ for both fits (Figs. 1–2, Table I). A focused test statistic (FTS) further tightens the 95% interval near the SM by 10–20% at no extra training cost (Fig. 3, Table II). **Completed: [✓ done]** #11, #14, #17. **Partial** (done without systematics, or by an alternative route): **[partial]** #1, #4, #7, #13, #15, #16, #18, #19, #20, #24, #25. Remaining items are systematics-, CR-, and data-driven-background-related.

Critical path

Statistics-first: port the JAX+IMINUIT fit (#11) and fix P_{ref} (#17), then build the in-situ-CR likelihood (#12) and the canonical ABCDnn flow with ExtendedABCD yields (#1,#2) in parallel; only *then* run Asimov closure (#13) and Neyman coverage (#14). The JES$\times\tau$ES/THU factorisation (#5), the coherent τES$\to$SVfit chain (#6), and the feature-ablation degeneracy study (#18) must land *before* coverage so the morphing is trustworthy; honest GP morphing-uncertainty (#9) is gated behind the MC-stat ensemble (#15). Multi-channel (#23), the k_{2V} tail (#21), POI-correlations (#22), and pipeline/pretraining (#25,#26) proceed as P1/P2 follow-ups.

Data-driven backgrounds

1. **[partial]** **[P0]** (L) **Canonical ABCDnn flow for QCD & jet$\to\tau$ fakes** (yield/shape split, three-piece uncertainty). MMD-trained NAF (continuous (N_j, N_b) conditioning, identity warm-up) morphs MC $\to$ the data-driven CR target; yield from ExtendedABCD/fake-factors, flow $\to$ per-event ratio $r_{\text{QCD}}(x)$ via a flow-vs-reference classifier; port the three four-top nuisances as per-event morphing; sweep target dim 2D$\to$5D$\to$10D. **Done:** VR closure $|1-\langle f\rangle| < 0.05$, RMS < 0.15; $r(x) \geq 0$ integrating to the ABCD yield; max stable dimension documented. *(Prelim.: fake-τ and QCD are already given a per-event density via parametrised mis-ID* weights *on* POWHEG*/MG5 MC; the ABCDnn flow + closure on real data remains.)* [§4]

2. **[P0]** (M) **ExtendedABCD vs standard-ABCD yield closure for $b\bar{b}\tau\tau$ QCD.** Standard ABCD failed ∼18–19% at high N_j in the four-top analysis and the QCD-norm NP is already large; test the extended estimator's closure across 6–8 orthogonal SRs binned in (N_j, N_b). **Done:** bias $< 7\%$, RMS $< 12\%$ per multiplicity bin; a safe-vs-extended region map. [§4]

3. **[P1]** (S) **Encode the one-sided QCD shape NP into smooth morphing.** CMS uses one alternative template for both up/down; test mirrored / one-sided-exponential / half-Gaussian encodings so the poly-exp morphing of $r_{\text{QCD}}(x)$ does not oscillate. **Done:** matches the CMS shape-impact rank ± 2, no oscillation, $< 2\%$ κ_λ bias. [§4]

4. **[partial]** **[P1]** (L) **Quantify the binned-QCD/fakes vs fully-unbinned cost.** Asimov ($\kappa_\lambda = 0, 1, 5$): full unbinned vs MC-only NSBI with QCD/fakes as a fixed binned template; also bin the NSBI score at 5/10/20/50 to check convergence. **Done:** width inflation $< 15\%$, bias $< 0.3\sigma$, monotonic convergence; a bin-vs-unbin fallback decision tree. [§4, §7]

Systematics & learned morphing

5. **[P0]** (M) **Validate factorisation & cross-terms: JES$\times\tau$ES and THU$_{HH}\times\kappa_\lambda$.** Both JES and τES feed m_{HH} (a cross-term absent in clean 4ℓ) and THU$_{HH}$ is POI-coupled. Test $r_{\text{comb}}/\prod_m r_m$ (median, KL) on held-out $\pm 1\sigma$ MC; compare naive-lnN vs κ_λ-coupled THU in Asimov. **Done:** factorisation residual 0.98–1.02, KL < 0.02; add a correction-NN if $> 2\%$. [§5]

6. **[P0]** (S) **Validate the coherent τES$\to$MET$\to$SVfit$\to m_{HH}$ recompute.** The weight-builder must shift τ energy at object level and recompute MET+SVfit+m_{HH} end-to-end, not shift m_{HH} directly. **Done:** matches the coherent truth to $< 2\%$ per-event MAE; the shortcut bias quantified. [§3, §5]

7. **[partial]** **[P1]** (M) **Validate poly-exp interpolation (κ_λ scan + $\pm 1\sigma$ anchors) under limited MC.** Intermediate κ_λ (0.5, 1.5) vs reweighted truth; downsample $\pm 1\sigma$ samples 100$\to$10%. **Done:** MAE $< 5\%$ down to ∼50% of MC, no gradient oscillation; minimum $\pm 1\sigma$ sample size to request from generators. [§1, §5]

8. **[P1]** (M) **Build & profile a CARL/DCTR per-event parton-shower/generator nuisance** (the NSBI upside). Train the per-event weight Pythia$\leftrightarrow$Herwig / ISR/FSR / VBF dipole-recoil; profile as one Gaussian NP vs the binned envelope. **Done:** KS $p > 0.1$, interval $\leq$ envelope with correct coverage, generator closure absorbed $< 2\sigma$. [§5]

9. **[P1]** (M) **Honest GP/ensemble-spread morphing-uncertainty with coverage calibration.** Fit a GP to per-event $r_{\rm true} - r_{\rm pred}$ residuals as a phase-space-dependent band (gated behind #15). **Done:** 65–70% coverage, calibration slope $1 \pm 10\%$, Spearman $\rho > 0.6$, narrower than a global band. [§5, §7]

10. **[P2]** (M) **Absorb b-tag/τ-ID SFs into the ratio, calibrated in situ** ($Z \to b\bar{b}$ / VZ($c\bar{c}$)). Demote the POG SF program to a cross-check and shrink the NN count. **Done:** in-situ SF within 30% of the POG prior, κ_λ not degraded, $|\Delta\text{pull}(0 \text{ vs } 5)| < 0.5\sigma$. [§5]

Statistics, fit & coverage

11. **[✓ done]** **[P0]** (M) **Port the jax+iminuit profile fit with the** $\{{\kappa_\lambda}^2, \kappa_\lambda, 1\}$ **basis** (root dependency). Replicate ATLAS Sec. 5: $P_{\rm ref}$, the signal/interference/background mixture with κ_λ-dependent rate, $\pm 1\sigma$ ratio classifiers + poly-exp, profile fit with Gaussian-constrained NPs; cross-check vs a binned Asimov. **Done:** converges, Hesse invertible, parabolic NLL; unbinned-vs-binned width $< 5\%$. *(Prelim.: unbinned extended likelihood [Eq. 3] with cached density ratios + profile-NLL scan [Fig. 1]; binned m_{HH} cross-check via `iminuit`. The NP-profiling framework for the ∼100 experimental systematics is the remaining piece.)* [§1, §6]

12. **[P0]** (M) **Represent the in-situ control regions (18 DY SFs, $t\bar{t}$-SF, QCD B/D) inside the unbinned likelihood.** Add Poisson CR auxiliary terms sharing the same NPs; compare SR-only-free-NP vs SR+explicit CRs. **Done:** reproduce the CMS in-situ SFs within 10–20%, no double-counting, corr(SF,κ_λ)< 0.3. [§4, §6]

13. **[partial]** **[P0]** (M) **Asimov & closure: morphing bias under injected shifts and generator/Delphes mismatch.** Inject known $\pm 1\sigma$ on JES/τES/QCD/MC-stat and re-fit; fit data from chain X with a model trained on Y (Pythia↔Herwig; Delphes↔full-sim). **Done:** recover κ_λ to < 0.05, pulls $\sim N(0, 1)$, sim-closure bias < 0.1 (flag full-sim retraining if $> 10\%$). *(Prelim.: Asimov + 500-toy closure done on **full-sim**; the DELPHES↔full-sim closure is the open piece, paired with sample production [#27].)* [§5, §6]

14. **[✓ done]** **[P0]** (M) **Neyman construction & coverage validation on MC-truth toys.** 1000+ toys; build the belt at $\kappa_\lambda \in \{-1, 0, 1, 3, 5\}$. **Done:** empirical coverage $\geq 67\%$, $|\text{bias}| < 0.1$ at SM; any failure traced to a specific NP/component. *(Prelim.: 500 toys at $\kappa_\lambda = 1$ give pulls $\sim \mathcal{N}(0, 1)$ and calibrated 68% intervals for both NSBI and binned [Fig. 2]; extend to the full κ_λ grid and re-run with systematics.)* [§6]

15. **[partial]** **[P1]** (M) **MC-stat ensemble-spread nuisance: design, size, coverage, Barlow–Beeston orthogonality.** Compare ensemble strategies (seeds / k-fold / bootstrap / m_{HH}-stratified) for the per-event ratio. **Done:** C2ST $p > 0.1$ on holdout, NP width ≈ 1 (BB-valid), κ_λ widens 5–15% when unfrozen, corr < 0.3 with physics NPs. [§5, §6]

16. **[partial]** **[P1]** (S) **Interference-aware impact definition + binned-NSBI fallback.** Implicit-function-theorem impact at the minimum vs finite-difference and the CMS rank; histogram the per-event log-ratio (5/10/20 bins) as an exact, learning-free cross-check. **Done:** matches the CMS rank ± 2; binned constraint $\geq 80\%$ of unbinned, monotonic convergence. [§6, §7]

Physics & POI

17. **[✓ done]** **[P0]** (S) **Define & validate the reference hypothesis $P_{\rm ref}$.** Compare SM / κ_λ-flat-mixture / pure-background / dedicated references; measure C2ST closure and how the $\{{\kappa_\lambda}^2, \kappa_\lambda, 1\}$ reconstruction degrades with $|\kappa_\lambda - \kappa_{\lambda\rm ref}|$, including the thin high-m_{HH} k_{2V} tail. **Done:** C2ST $p > 0.05$ across the scan, $|\log r| < 10$, $< 5\%$ basis bias at extremes. *(Prelim.: a single distant reference $\kappa_\lambda = 5$ was chosen; a pooled-mixture reference gave 0.50 classifier accuracy [target–reference self-overlap] and was rejected; $r_{\rm ref} \equiv 1$; stable against reference-sample replacement.)* [§1]

18. **[partial]** **[P0]** (M) **Feature-ablation degeneracy-breaking: $\rho(\kappa_\lambda$,JES/τES) and VIF vs feature schedule.** $m_{HH}\to$10-feature$\to$ +constituents$\to$ +SVfit covariance$\to$ +resolution handles; ablate to find the critical features and the asymptotic ρ floor. **Done:** reproduce $\rho\, 1.000 \to 0.646$; a minimal set with $\rho < 0.7$, VIF< 1.5; settle whether constituents/SVfit push $\rho < 0.60$ or it plateaus. *(Prelim.: the* degeneracy-break is demonstrated *— the 10-feature NSBI lifts the binned-m_{HH} secondary minimum near $\kappa_\lambda \sim 3$–4 [Fig. 1]; features selected by total-variation distance + integral-closure, stable under one-at-a-time low-TV removal. The $\rho(\kappa_\lambda$,JES/τES)/VIF quantification needs the systematic variations.)* [§2, §5]

19. **[partial]** **[P1]** (S) **Source the κ_λ-differential THU$_{HH}$ band — and measure the shape-truncation bias.** Compile the scale/PDF/m_t band vs $\kappa_\lambda \in [-2, 5]$ and parametrise it as a κ_λ-dependent *rate* weight; then fit Asimov data generated with scale-varied *shapes* using a normalisation-only model and measure the κ_λ bias. **Done:** a smooth band reproducing the AN-25-103 Table-40 values (incl. the m_t sign flip); truncation bias quantified ($< 0.05 \Rightarrow$ CMS-style normalisation-only is justified). *(Prelim.: the total-level band per κ_λ now exists — AN-25-103 Table 40: $+4/-18\%$ at κ_λ=1, $+4/-22\%$ at 2.45, $+13/-4\%$ at 5 [m_t scheme]; the* component-differential *split needed for shape morphing remains open — C4b, see §5.)* [§2, §5]

20. **[partial]** **[P1]** (M) **Validate the LO→NLO κ_λ-reweighting that generates the $R(x|\kappa_\lambda)$ sources.** Direct-NLO vs 2D-LO-reweight vs dedicated-NLO-reweight; propagate the difference through an Asimov fit. **Done:** NLO-reweight reproduces direct to $< 5\%$/bin and $< 0.1\sigma$ κ_λ bias; the LO-only cost quantified. *(Prelim.: sidestepped — signal generated* directly *at NLO [POWHEG] at $\kappa_\lambda \in \{0, 1, 5\}$ with a 3-point quadratic basis [the $\kappa_\lambda = 2.45$ point was dropped for morphing stability]; this study still applies if the CMS LO+reweighting samples are used instead.)* [§2]

21. **[P1]** (M) **Measure k_{2V} sensitivity in the high-m_{HH} VBF tail via unbinned NSBI.** VBF-enriched ($M_{jj} > 400$ GeV), k_{2V}-morphed MC; Asimov-fit binned vs unbinned vs hybrid; cross-validate the tail (train low-m_{HH}, test high). **Done:** NSBI k_{2V} error 20–40% smaller, tail-extrapolation bias $< 10\%$, a minimum viable bin granularity. [§2]

22. **[P1]** (L) **Map the rate-broken POI correlations ($\kappa_t \leftrightarrow \kappa_\lambda$, $k_V \leftrightarrow k_{2V}$) and confirm the scope caveat.** Multi-POI grid; profile-likelihood correlation contours; attribute information to shape vs rate. **Done:** a POI correlation matrix; NSBI lowers $\rho(\kappa_\lambda$,$k_{2V})$ but does *not* improve the rate-broken pairs. [§2, §7]

23. **[P1]** (M) **Build the multi-channel + boosted simultaneous unbinned likelihood.** Per-channel vs joint morphing for the three $\tau\tau$ channels + boosted, with the CMS year/channel NP-correlation scheme. **Done:** combined κ_λ width within 5% of the best combination; no extra $> 0.5\sigma$ pulls. [§1, §3]

ML, samples & pipeline

24. **[partial]** **[P1]** (M) **Low-signal robustness: morphing calibration & uncertainty honesty at $\mathcal{O}$(10–100) signal events.** Downsample the $\pm 1\sigma$ sets and effective signal yield (50/100/250) and retrain per level. **Done:** C2ST $p > 0.05$ at the $\pm 1\sigma$ floor, ensemble spread $< 2\times$ the high-stat value; flag the binned fallback if it fails. [§5, §7]

25. **[partial]** [P2] (L) **Reproducible Perlmutter pipeline for the ∼2000-NN morphing ensemble.** Containerise Sophon+JAX+IMINUIT, a training manifest, resume-on-failure; benchmark per-NP classifiers vs a single conditional morphing network. **Done:** 500 NNs in < 4 h (∼1 GPU-day), $> 99\%$ resume, bit-identical reruns; the per-NP-vs-conditional trade-off documented. [general]

26. [P2] (M) **Pretraining follow-up: signal-vs-$t\bar{t}$ foundation-model benchmark vs scratch** (confirmatory). Scratch vs $t\bar{t}$-kinematics-pretrain vs HH-kinematics-pretrain; F-score κ_λ= 0 vs 5 by m_{HH} region. **Done:** signal-vs-$t\bar{t}$ pretraining closes $\geq 30\%$ of the FM deficit, else confirm scratch remains the baseline. [established null]

27. **[partial]** **[P1]** (M) **Produce & validate delphes $b\bar{b}\tau\tau$ samples for the phenomenological-paper version.** The sensitivity is currently validated on **CMS full-simulation** samples; a publishable

pheno paper needs a DELPHES fast-sim reproduction. Validate the tuned CMS DELPHES card (b-tag, τ-ID/DeepTau, MET, lepton response) against the full-sim object performance and the reconstructed-vs-generated m_{HH} response, then regenerate signal + backgrounds and re-fit. **Done:** DELPHES-vs-full-sim closure on key observables (especially m_{HH} below 400 GeV) within the per-event-weight tolerance; the κ_λ interval reproduced to $< 0.1\sigma$. *(Prelim.: a standalone DELPHES card-validation pipeline is built; full sample production for the pheno paper is in progress. Pairs with #13.)* [§3, general]

1 The two measurements and the shared NSBI skeleton

ATLAS off-shell [ATLAS p. 1,5]: measures the off-shell signal strength $\mu_{\text{off-shell}}$ (hence the Higgs width) in $gg \to (H^* \to)ZZ \to 4\ell$, using $139\,\text{fb}^{-1}$. The physics is a *coupling that morphs an interference shape* (triangle–box in $ggZZ$). **CMS HH→ $b\bar{b}\tau\tau$** [CMS p. 3,4]: searches for non-resonant HH via ggF+VBF in $b\bar{b}\tau\tau$ ($\mathcal{B} = 7.3\%$), $137.6\,\text{fb}^{-1}$, three $\tau\tau$ channels ($\tau_e\tau_h, \tau_\mu\tau_h, \tau_h\tau_h$, plus boosted), extracting κ_λ (and k_{2V}, k_V). $\sigma^{HH}_{ggF} = 31.05^{+2.2}_{-5.0}\%$ fb, $\sigma^{HH}_{VBF} = 1.726$ fb [CMS p. 4].
What transfers (the skeleton). The ATLAS recipe (Sec. 5) is reusable wholesale: a reference-hypothesis P_{ref} so all NNs estimate ratios to a fixed point; a search-oriented *mixture model* for signal/interference/background; nuisance parameters in both the rate and the per-event density; a profile-likelihood test statistic in a custom JAX+IMINUIT fit; and an interference-aware impact definition [ATLAS p. 89,90]. The off-shell interference structure $(\mu - \sqrt{\mu})\, S + \sqrt{\mu}\,\text{SBI} + (1 - \sqrt{\mu})\, B$ [ATLAS p. 81] maps onto the κ_λ morphing basis $\{\kappa_\lambda^2, \kappa_\lambda, 1\}$ (since $gg \to HH = \kappa_\lambda \cdot \text{triangle} + \text{box}$). *The challenges below are about the inputs to this skeleton, not the skeleton itself.*

2 Physics / parameter-of-interest challenges

- **The POI lives in a shape, and that shape is m_{HH}.** κ_λ sensitivity is concentrated in the m_{HH} spectrum (triangle–box interference), strongest in the resolved regime $m_{HH} \in [250, 400]$ GeV. CMS encodes the κ_λ dependence by *reweighting* the signal MC — a 2D $(m_{HH}, |\cos\theta^*|)$ LO reweighting and a dedicated NLO reweighting for the final extraction [CMS p. 6]. In NSBI this becomes the $R(x|\kappa_\lambda)$ factor.
- **Multiple correlated POIs / EFT directions.** Beyond κ_λ the analysis also probes k_{2V}, k_V and HEFT operators (c_2, c_{2g}, c_g) [CMS p. 4]; the off-shell measurement is effectively one POI. More POIs $\Rightarrow$ a higher-dimensional morphing.
- **k_{2V} sensitivity lives in the high-m_{HH} VBF tail — where binning hurts most.** The $HHVV$ contact term grows with energy unless $k_{2V} = 1$, so VBF-HH k_{2V} sensitivity is concentrated in the high-m_{HH} tail; current analyses use only ∼4–5 bins there. This is precisely the regime where coarse binning discards the most information, so the unbinned NSBI gain is largest on the k_{2V} (and the low-m_{HH} κ_λ) shape directions. (k_{2V} is now the best-measured “inaccessible” coupling — the 2026 ATLAS+CMS combination gives $k_{2V} \in [0.73, 1.3]$.)
- **Interference impact ambiguity (shared, but worse).** Both have interference, so the finite-difference NP-impact definition is ambiguous; ATLAS replaces it with a local implicit-function-theorem impact [ATLAS p. 90]. κ_λ needs the same fix.

3 Final state and reconstruction

The off-shell 4ℓ final state is *exquisitely clean*: four leptons, fully reconstructed, with tiny energy-scale systematics and no jets/MET/τ/b-tagging in the core. $b\bar{b}\tau\tau$ is the opposite:

- **b-jets** (JES/JER, b-tagging) + **hadronic τ** (τ-ID, τ energy scale) + **MET** (the τ neutrinos). The $\tau\tau$ mass is *not* fully reconstructed — it requires an algorithm (SVfit); shifting τ energy shifts MET *and* SVfit coherently [CMS p. 93].
- **Three channels** with different background compositions and systematics, plus a boosted regime.

Consequence: the observable x for $b\bar{b}\tau\tau$ is softer and carries many more systematic "handles" than 4ℓ.

4 Background handling — the largest qualitative difference

The crux

Off-shell backgrounds are **all MC-based and clean** (one dominant irreducible $q\bar{q}ZZ$ continuum, modelled in simulation, plus EW). $b\bar{b}\tau\tau$ backgrounds are a **zoo, several of them data-driven**, and *data-driven backgrounds have no per-event MC probability density* — which is the object NSBI needs. This is the single hardest conceptual mismatch between $b\bar{b}\tau\tau$ and the off-shell NSBI template.

ATLAS off-shell (Sec. 2.5, Sec. 5). The dominant background is the irreducible $q\bar{q} \to ZZ \to 4\ell$ continuum, which *interferes* with the signal and is carried in the mixture model as a component $\mu_{q\bar{q}ZZ}$ (three floating NPs) constrained *in situ* by a dedicated n_{jets} discriminant in a control region [ATLAS p. 81]. Plus a small EW component. Everything is MC, so every process has a per-event density the NSBI ratios can target.

CMS $b\bar{b}\tau\tau$ (Sec. 7). A heterogeneous set, with the two largest estimated *from data*:

- **QCD multijet — data-driven ABCD** [CMS p. 61,62]. Four regions in (charge $\times$ τ-isolation); shape from region C (data minus other-MC), yield from $C \times B/D$. The QCD *normalisation* uncertainty ranges **5–100%** by category (Table 30, [CMS p. 92]); its *shape* uncertainty (alternative B-vs-C templates, symmetrised) is **one of the highest-ranking nuisances** in the most sensitive $\tau_h\tau_h$ channel [CMS p. 91,96,97].
- **Drell–Yan (Z+jets) — MC normalised by 18 data-driven scale factors** [CMS p. 63]. A simultaneous fit of $M(\mu\mu)$ across **18 control regions** (3 b-jet multiplicities $\times$ 6 Z-p_{T} bins), giving 18 normalisation SFs ($\sim$0.6–1.7) *and* 18 shape templates (scaled $\pm$) propagated with their covariance [CMS p. 65,94].
- **$t\bar{t}$ — MC shape + data-driven normalisation SF** [CMS p. 79]. A dedicated CR (inverted elliptical (m_{HH}) mass cut, Eq. 22) with $< 7\%$ contamination; COMBINE fit with Barlow–Beeston gives ttSF $\approx$ 0.91–0.99 [CMS p. 81,82]; the shape is taken from MC (no shape dependence observed).
- **Jet$\to$ τ fakes — data-driven** [CMS p. 94,95]. $> 93\%$ of the events with a jet faking a τ; fake-rate uncertainty per region (barrel 24/13/11%, endcap 28/18/25% by year, Table 33), applied as an event weight.
- **Single Higgs — irreducible, same final state** [CMS p. 82]. ggH, VBFH, VH, $t\bar{t}H$ with $H \to b\bar{b}$ or $\tau\tau$ mimic the signal; modelled from MC. (Note: $t\bar{t}H/VH$ couplings make this a *coupling-dependent* background.)
- **Di-/tri-boson, W+jets, single-t, V+2jets — MC** [CMS p. 82].

Why this is a new, NSBI-specific difficulty

1. **No density for data-driven backgrounds.** QCD (ABCD) and jet$\to\tau$ fakes are estimated as *yields/weights from data control regions*, not as simulated events with a per-event probability. NSBI's per-event likelihood ratio $P_X(x)/P_{\text{ref}}(x)$ is undefined for them. Folding an ABCD/fake estimate into an *unbinned* per-event likelihood is an open problem the off-shell analysis never had to solve (its one big background was MC). Options (all imperfect): treat QCD/fakes only in a binned auxiliary term; build a surrogate density from the control-region data; or restrict NSBI to the MC-modelled part and bin the rest.

2. **The QCD shape systematic is a top-ranked nuisance** [CMS p. 91] — and it is intrinsically *asymmetric/ill-defined* (one alternative template used for both up and down), forcing a symmetrisation hack [CMS p. 96,97]. Carrying that into a per-event morphing is non-trivial.

3. **Single Higgs is an irreducible, coupling-dependent background** in the same final state — it must be modelled *and* morphed consistently with the signal couplings, with its own (correlated) theory systematics.

4. **Many background-normalisation NPs are themselves data-fit results** (18 DY SFs with a covariance, ttSF, QCD B/D) — these are constrained *in situ* in the binned CMS fit; reproducing that constraint inside an unbinned NSBI fit requires the CRs to be represented in the per-event likelihood too.

A workaround: flow-based data-driven backgrounds (ABCDnn)

The "no per-event density" problem is the sharpest objection to NSBI here, but it is *not* a dead end — there is a demonstrated solution. **ABCDnn** (Choi, Lim, Oh, arXiv:2008.03636) generalises the ABCD method with a *neural autoregressive flow* that learns the full multi-dimensional CR$\to$SR transformation, conditioned on the discriminating observables. Where standard ABCD gives only a scalar SR yield ($N_D = N_B N_C/N_A$), ABCDnn gives a full *differential* background distribution $p_{\text{QCD}}(x)$ — which can then be turned into a density ratio $r_{\text{QCD}}(x) = p_{\text{QCD}}(x)/p_{\text{ref}}(x)$ by training a classifier between flow-generated events and the pooled reference, so it enters the NSBI likelihood *on the same footing* as the MC-derived ratios.

The concrete production example (S. An, Dutta/Paulini, CMU PhD thesis 2025, Ch. 5): the CMS *all-hadronic four-top* search, Run-2 ($137\,\text{fb}^{-1}$, 13 TeV) + Run-3 ($62\,\text{fb}^{-1}$, 13.6 TeV), reaching an **observed (expected) significance of** $3.95\,(1.65)\,\sigma$, $\mu = 2.7^{+1.1}_{-0.9}$. Its single most important design choice is a **yield/shape split** of the dominant $t\bar{t}$+jets and QCD backgrounds:

- *Yield* $\leftarrow$ **ExtendedABCD**: standard ABCD in (N_{jets}, N_b) *fails* here (the factorisation breaks at high multiplicity — off by ∼18–19.5% in closure); adding two extra sidebands restores it to ∼1–7% (their $\hat{F}_D = (F_B F_C/F_A)^2(F_X/F_B F_Y)$, Table 5.1–5.2). The lesson: *get the normalisation from a robust closed-form estimator with enough sidebands, not from the flow.*

- *Shape* $\leftarrow$ **ABCDnn** (a neural autoregressive flow): it learns a transformation that *morphs the $t\bar{t}$ MC into the data-driven $t\bar{t}$+QCD estimate* (target = data − signal − minor bkg per CR), trained by minimising **Maximum Mean Discrepancy** and *conditioned on the CR position* (N_{jets}, N_b), then applied to $t\bar{t}$ MC in the SR. Notably it never uses QCD MC (too few SR events) — the flow turns pure-$t\bar{t}$ MC into an effective $t\bar{t}$+QCD mixture.

Three training tricks they found necessary (worth copying): **loss-weighted CR sampling** (sample harder CRs more often, $\propto e^{\text{loss}}$, so every region is learned uniformly); **continuous** (N_{jets}, N_b) **conditioning** instead of one-hot labels, so the flow *interpolates smoothly* across CRs — this is what makes the CR→SR *extrapolation* well-behaved; and a **warm-up** (pre-train near identity, RealNVP-style) for stable convergence.

Application to $b\bar{b}\tau\tau$: the **jet→ τ fakes** are the natural target — a fake-factor method is conceptually an ABCD with τ-ID inversion as the second axis, so an ABCDnn conditioned on $(p_{\mathrm{T}}^{\tau}, \eta^{\tau})$ could learn the full fake-τ kinematic shape instead of the current coarse 2D fake-factor binning; the QCD multijet is the other target; and for $t\bar{t}$ the flow can serve as a data-driven shape cross-check. Following the four-top lesson, keep the **yield/shape split**: take the fake-τ/QCD *normalisation* from the existing fake-factor/ABCD machinery (which CMS already trusts, [CMS p. 61,94]) and use ABCDnn only for the differential *shape*, then convert that shape to a per-event ratio $r(x)$ for the likelihood. A fully realised $b\bar{b}\tau\tau$-NSBI would use *ABCDnn shapes (with ABCD/fake-factor yields) for QCD/fakes, MC ratios for the MC-modelled components (Z+jets, $t\bar{t}$, single-H, diboson), and one unbinned NSBI likelihood combining both* — removing the main caveat that NSBI "depends on MC where MC is wrong."

But the generic caveats remain: (i) *extrapolation distance* — the CR→SR map strains for tight SRs far from the sidebands in tagger/isolation space (multiple overlapping CRs validate it but cost stats); (ii) *uncertainty propagation* — the four-top analysis does *not* bootstrap the flow; it splits the uncertainty into three data-driven nuisances (thesis Sec. 6.1, Table 6.1): the *finite-CR-statistics* piece is a lnN normalisation NP propagated through the ABCD formula ("statistics of extended-ABCD," 4–26%); the *normalisation-modelling* piece is a lnN NP from a validation-region (VR) data-vs-prediction closure (weighted-mean deviation $1 - \langle f \rangle$ + weighted RMS of $f = N_{\mathrm{data}}/N_{\mathrm{pred}}$, 6–44%); and the *ABCDnn shape* uncertainty is a shape NP implemented as a *linear shift of the discriminant*, with the shift size chosen by *minimising a quantitative data/pred-agreement metric* in the VR (floor 1%, correlated across H_T, uncorrelated across years). The whole uncertainty-*estimation* procedure is itself validated in a dedicated orthogonal "closure-test region" that checks pred+syst covers truth in the SR. *Caveat for NSBI:* this VR-closure shape nuisance ("shift the discriminant") is a *binned-fit* construct; in an unbinned NSBI it becomes yet another per-event morphing nuisance on the ratio $r(x)$ — i.e. the same learned-morphing problem, now applied to the data-driven component; (iii) *signal contamination in CRs* — if signal leaks into the control region the flow learns "background + signal" and biases the SR low, requiring iterative CR definitions with signal subtraction (this caveat is *mild* for $b\bar{b}\tau\tau$: the HH signal is tiny in the fake-τ/QCD control regions, just as the SM four-top signal was negligible in its CRs); (iv) *no theory handle* — the flow gives a shape but no ME+PS+PDF decomposition, so generator-level theory uncertainties on the background cannot be propagated and are replaced by a data-driven CR→SR closure systematic.

The one caveat specific to using ABCDnn *for NSBI* (the real gap). In the four-top thesis the flow models only a *low-dimensional* target — the event-level BDT discriminant plus H_T (two variables) — because the downstream fit is a *binned template* fit of that discriminant. NSBI needs the opposite: the *full per-event density* over the whole feature vector, so the data-driven shape can become a per-event ratio $r(x)$. So the four-top result proves flows can deliver a data-driven background *shape of a summary observable*; extending that to the full-dimensional per-event density an unbinned NSBI consumes is a genuine step beyond what is demonstrated — a higher-dimensional flow, harder to train and validate. The conservative intermediate is to let ABCDnn produce the data-driven shape of the *NSBI discriminant* (the per-event log-ratio summed over components), i.e. treat QCD/fakes as a binned component of an otherwise-unbinned likelihood — exactly the "keep QCD binned, NSBI-fy the MC components" compromise.

Net: ABCDnn downgrades the data-driven-background problem from "open" to "solved-with-caveats" for a binned *discriminant shape; the fully-unbinned version is a worthwhile but unproven sub-project for a $b\bar{b}\tau\tau$-NSBI pilot.*

5 Systematics handling

What is genuinely shared (equal for both, and equally fallible): the *sources and* $\pm 1\sigma$ *sizes* of the NPs; the *object-level propagation* (shift the object and all dependents — e.g. $\tau \Rightarrow$ MET$\Rightarrow$SVfit — and recompute the discriminant, [CMS p. 93]); the *factorisation over NPs* and $\pm 1\sigma$ *interpolation*; *Gaussian-constrained profiling*; and the κ_λ*-dependent theory* treatment. ATLAS builds the per-event systematic weight as $P_X(x,\alpha) = w_X(x,\alpha)\,P_X(x,0)$ with $w_X(x,\alpha) = \prod_m w_X(x,\alpha_m)$, each single-NP weight interpolated poly-exp between $\pm 1\sigma$ classifier estimates [ATLAS p. 83,84]. CMS does the per-*bin* analogue (re-run the DNN on varied inputs $\to$ template).

What is harder for $\kappa_\lambda/b\bar{b}\tau\tau$:

1. **κ_λ–systematic shape degeneracy.** The dominant experimental systematics (JES, τES, MET, signal parton shower) deform *the same* m_{HH} *shape* that carries κ_λ. In 4ℓ the big systematics (theory) are largely *separable* from the signal shape; in $b\bar{b}\tau\tau$ they are not. Decorrelation is therefore impossible (it would erase the signal), forcing the "model-it" route — so the accuracy of the morphing $\Delta(x|\alpha)$ *directly sets* the κ_λ uncertainty. (A Fisher-information toy gives, schematically, a $\sim$13$\times$ systematic inflation from m_{HH} alone collapsing to $\sim$1.3$\times$ with the full feature set — the degeneracy is broken by dimensionality, not avoided.)

2. **An order of magnitude more, and larger, experimental NPs** [CMS p. 90–97]: JES split into a reduced set of **11 sources** with a year-correlation scheme [CMS p. 95]; τ**ES** p_{T}-dependent with 4 decay-mode components [CMS p. 94]; ***b*-tag** 5 components (hf, lf, hfstats, lfstats, cferr) [CMS p. 96]; **DeepTau** vsJet (5 p_{T} bins) + vsEle (2 η bins) [CMS p. 95,96]; trigger SFs, JER, PU-jet-ID, L1 prefiring, $e/\mu \to \tau$ fake ES. The off-shell analysis has comparatively few, small (lepton) experimental NPs. Each NP is another per-event weight to learn ($\sim$2000 NNs already in off-shell = 2$\times$NP$\times$process; $b\bar{b}\tau\tau$ grows this).

3. **Data-driven background systematics that don't fit the density-ratio picture**: the QCD norm (5–100%) and high-ranking QCD shape [CMS p. 91,92,96,97], the 18 DY SF shape/norm templates [CMS p. 94], the jet$\to\tau$ fake-rate uncertainties (11–28%, Table 33) [CMS p. 95], the VBF dipole-recoil PS uncertainty (20–70%, Table 32) [CMS p. 93]. These are template/weight constructs, not deformations of an MC density.

4. **κ_λ-dependent theory uncertainty (POI$\times$systematic entanglement).** The HH scale/PDF/m_t uncertainties are folded into normalisation NPs that are strongly κ_λ*-dependent*: Run-2 quoted `THU_HH` $= +4\%/-18\%$ at the SM [CMS p. 91]; the Run-3 note (AN-25-103) gives ggF scale+m_t-scheme $= +6/-23\%$ at the SM and tabulates the band *per* κ_λ *point*, with the m_t asymmetry *flipping sign* across the scan (its Table 40). The off-shell μ does not entangle with its theory uncertainty this way. A faithful *per-event* treatment would require POI-coupled morphing — but see the AN-25-103 reassessment below: the *rate* part is handled pointwise without any such object; only the *shape* part remains open.

5. **Factorisation is more likely to break.** $w_X = \prod_m w_X(\alpha_m)$ assumes NPs are independent sources [ATLAS p. 83]; but JES and τES *both* feed m_{HH} $(= m(b\bar{b}+\tau\tau))$, so a genuine cross-term exists in $b\bar{b}\tau\tau$ that is benign in clean 4ℓ.

6. **MC-statistical uncertainty of the learned ratios.** ATLAS propagates it as one NP via the NN-ensemble spread [ATLAS p. 86,87]; CMS uses Barlow–Beeston per bin. With far smaller *systematic-variation* samples in $b\bar{b}\tau\tau$, this term is more important (the nominal MC, $\approx$500k–750k/κ_λ-point, is ample for training; the binding constraint is the least-populated $\pm 1\sigma$ sample).

Candidate estimators (the choice to settle before quoting a number): the ATLAS NN-ensemble-spread/double-bootstrap (argued *incorrect* by Shyamsundar, 2505.19156); and the analytic-frequentist **wifi ensembles** (Benevedes & Thaler, 2506.00113) — a basis-function expansion whose weight covariance propagates in closed form to a per-event uncertainty band on $r(x)$, with demonstrated coverage on an SBI density-ratio example and *no bootstrap*, so it is attractive on both correctness *and* the ensemble/compute cost (untested on interference/morphing/weighted-MC).

Two systematics that NSBI actually handles *better* than templates. Worth stating the upside too: (a) *parton-shower / generator* uncertainties (Pythia vs. Herwig, ISR/FSR) are a natural fit — a **CARL/DCTR**-style reweighting network learns the per-event weight between alternative showers, which is then profiled as a nuisance; this is strictly more powerful than the current "shape envelope over generator variations" and is the in-likelihood cousin of the RS3L robust-representation idea. (b) b-tag / τ-ID scale factors can be *absorbed* into the density ratio and calibrated in situ against a standard candle (the $VZ(c\bar{c})/Z \to b\bar{b}$ peak trick), turning the POG SF program into an external cross-check rather than the primary constraint. So the systematics ledger is not all cost: *the morphing-heavy ones (JES/τES on m_{HH}) are the burden; the modelling ones (shower, flavour-tagging) are where NSBI's per-event treatment is an asset.*

Literature maturity & risk (where each challenge actually stands). A targeted sweep of the 2024–2026 literature shows that *none* of the six difficulties above is cleanly "numerical" (method exists, just measure how bad): the $b\bar{b}\tau\tau$ regime consistently exceeds where any per-event method has been demonstrated. The largest published demonstrations of *learned* NP morphing are $\sim$3 calibration NPs on *simulated* data (GOLLUM; Schöfbeck *et al.*, PRD 112 052006) and $\sim$50 NPs on real but *clean* 4ℓ data (ATLAS, 2412.01600). Nothing exercises the actual $b\bar{b}\tau\tau$ regime: $\sim$40 *shape-degenerate* experimental NPs deforming a *reconstructed* m_{HH} across three channels with data-driven backgrounds.

Challenge (item above)	Literature maturity	Class	Risk
1 κ_λ–syst. shape degeneracy	poly-exp morphing exists (ATLAS, GOLLUM); per-event closure under *simultaneous* JES$\times\tau$ES on m_{HH} unreported	hybrid	high
2 order-of-mag. more NPs ($\sim$40)	$\leq$50 NPs on clean 4ℓ (ATLAS); $\leq$3 calib. NPs on sim data (GOLLUM); cross-terms + calibration at this scale unproven	hybrid	high
3 data-driven bkg systematics	components exist (ABCDnn, neural FF, GOLLUM) but *never combined* per-event; neural FF (2511.06972) explicitly defers systematics	hybrid (borderline open)	high
4 κ_λ-dependent (POI-coupled) theory	*split* (AN-25-103): *rate* band = standard practice (pointwise κ_λ-dependent lnN, its Table 40); per-event *shape* part = proposed only, done by nobody	**C4a closed / C4b open (upgrade)**	med
5 factorisation cross-term (NP$\times$NP)	per-event *toy* only (GOLLUM, sim data); τES$\to$MET$\to$SVfit$\to m_{HH}$ recompute untested	hybrid	med
6 MC-stat of learned ratios	real (ATLAS) but a *published correctness critique* (Shyamsundar, 2505.19156); leading candidate fix = analytic-frequentist *wifi ensembles* (2506.00113)	hybrid	high
a PS/generator via CARL/DCTR	components exist; never demonstrated end-to-end as a profiled NSBI nuisance	hybrid	med
b absorb b-tag/τ-ID SFs	building blocks only; no per-event SF-absorption + in-situ standard candle shown	hybrid	med

Open vs. numerical — the verdict

One challenge was rated conceptually open — and it has since split (see the AN-25-103 reassessment below): #4, the κ_λ-dependent / POI-coupled theory morphing $w_\theta(x;\kappa_\lambda,\alpha_\theta)$. The standard factorised scheme keeps every NP weight *POI-independent* ($w_m(x;\alpha_m)$), and off-shell avoids the problem entirely (μ does not couple to its theory band because its signal *shape* is a fixed mixture). Note #4 (POI$\times$NP) is distinct from #5 (NP$\times$NP): GOLLUM solves the latter on a toy but does *not* reach the former. *Post-AN-25-103*: the *rate* half (C4a) is **closed** — a pointwise κ_λ-dependent lnN, standard CMS practice — while the per-event *shape* half (C4b) remains open but is an *upgrade* that no analysis performs. #3 (per-event systematics of a *data-driven* density) is the borderline second — every component exists but none combine. The other six are *hybrid*: a method exists in an easier setting, so each "pilot" is a *method-selection gate*, not pure quantification. **Most dangerous to the headline** (a *tighter* interval that must still *cover*): #6 — Shyamsundar (2505.19156) argues via a toy that the ATLAS double-bootstrap MC-stat estimator does *not* capture the finite-MC uncertainty, so settle this before quoting a number (leading candidate replacement: the analytic-frequentist *wifi*-ensemble estimator, 2506.00113); and #2 — the $\sim$40-NP scale may force a conditional surrogate (GOLLUM/R4) or GP morphing (R3) over $\sim$thousands of classifiers. **Suggested gates first:** #4 (does a POI$\times$NP ansatz work, on a toy) and #6 (which MC-stat estimator is correct); then the #2 scaling gate; then #3/#1/#5 closures; the upsides (a,b) are field-first demonstrations. *(Gate #4 refocused post-AN-25-103: the rate half is closed; the pilot is now the theory-*shape *truncation-bias measurement — see the reassessment below.)*

Why off-shell escapes #4, and the likely way out. Algebraically off-shell and κ_λ are *identical* — POI coefficients on fixed templates, $(\mu-\sqrt{\mu},\ \sqrt{\mu},\ 1-\sqrt{\mu})$ vs. $(\kappa_\lambda^2,\ \kappa_\lambda,\ 1)$ — so the difference is not the POI structure but *what the theory uncertainty acts on.* THU_{HH} sits on the *signal*, and the triangle/box/interference carry *different* scale/PDF/m_t bands; κ_λ sets their mixture, so the net theory deformation's size *and* its m_{HH} shape vary with κ_λ (the *fractional* band even diverges near the $\kappa_\lambda \approx 2.45$ cross-section minimum, where triangle and box cancel). In off-shell the leading theory NP is the $gg \to ZZ$ continuum K-factor — on the *background* (coefficient 1, μ-independent) — and σ is monotonic in μ with no cancellation, so μ never re-mixes a component-differential theory band. **Way out (first thing to try):** replace the single total-signal THU_{HH} NP with *per-component, POI-independent* theory weights $w_\triangle$, w_{int}, $w_\Box$ (functions of α_θ only); the κ_λ coefficients then blend them and the κ_λ-dependence of the net THU emerges *automatically*, preserving factorisation. The two missing pieces (hence "open"): the *component-differential* theory bands (the WG tabulates only the total band per κ_λ point), and any NSBI closure test of per-component theory morphing. *This prediction is now empirically confirmed — see the sign flip in the AN-25-103 update just below.*

Update (12 Jul 2026): how the theory uncertainty is *actually* handled in the current CMS $b\bar{b}\tau\tau$ analysis (Run-3, AN-25-103) — and the C4 reassessment. Reading the Run-3 note confirms the theory uncertainty is the *leading* $b\bar{b}\tau\tau$ systematic and reveals a three-layer treatment:

1. *POI rate+shape, by construction.* The ggF signal at any $(\kappa_\lambda, \kappa_t)$ is an *exact* three-component template combination, $\sigma(\kappa_\lambda, \kappa_t) = \mathbf{k}^{\mathrm{T}}\mathbf{K}^{-1}\boldsymbol{\Sigma}$, built from three NLO POWHEG samples at $\kappa_\lambda =$ 1, 2.45, 5 and applied *per bin at fit time* (AN Eq. 52–55; the VBF analog uses six components). This is precisely the mixture / known-green-coefficient structure of Sec. 1, executed on templates.

2. *Theory NPs are pure normalisation lnN.* QCDscale, PDF and α_s enter as rate modifiers with log-normal priors (AN Sec. 7.1, Table 42): at the SM point, ggF scale+m_t-scheme $= +6/-23\%$ (the -23% dominated by the m_t*-scheme* choice, not scale), PDF $\pm1.5\%$, α_s $\pm1.7\%$; VBF scale $+0.05/-0.03\%$, PDF+α_s $\pm2.7\%$. **No theory-shape NP exists for the signal** — scale/m_t variations never deform the templates.

3. *The κ_λ-dependence is handled at interpretation time.* AN Table 40 tabulates the band *per κ_λ basis point* — large and *sign-flipping*:

κ_λ	$\sigma(\kappa_\lambda)$ [fb]	scale	m_t scheme
1	34.13	$+2.1/-4.9\%$	$+4/-18\%$
2.45	14.92	$+3.2/-5.5\%$	$+4/-22\%$
5	99.65	$+7.9/-8.4\%$	$+\mathbf{13}/-\mathbf{4}\%$

   The scans are *pointwise*: at each fixed κ_λ hypothesis the model is *rebuilt* (templates via Eq. 55, band at that κ_λ) and all nuisances profiled (AN Sec. 8.4.5); 2-D exclusions compare the rate limit to σ_{theory} *evaluated at each parameter point*, with the band propagated to the exclusion boundaries (Sec. 8.4.6); the Run-2+3 combination treats the joint QCD+m_t ggF uncertainty as *correlated* across runs (Sec. 8.5.1).

Concept: what a "scan" actually is — and why κ_λ can be fixed inside every fit

A profile-likelihood "scan" is not one fit: it is a **grid of conditional fits, one per hypothesis value**. Each fit answers one frozen question — *"if κ_λ were exactly this value, how well can the model explain the data, with every nuisance doing its best?"* — and returns *one number*, the profiled NLL: one evaluation of the numerator of $\lambda(\kappa_\lambda) = L(\kappa_\lambda, \hat{\hat{\alpha}}(\kappa_\lambda))/L(\hat{\kappa_\lambda}, \hat{\alpha})$. The "scan" is the assembled table of (κ_λ, t) pairs; $\hat{\kappa_\lambda}$ is the *curve's* minimum and the CI its $t \leq 1$ crossings — read off the assembly, never from a single fit. (Valley picture: each fit is a probe dropped to the valley floor at one longitude; no probe moves sideways, yet the probe depths trace the floor-vs-longitude curve.) **Why pointwise** rather than one fit with κ_λ floating: (i) a free fit yields only the minimum and a Hessian *parabola* — blind to the grossly non-parabolic κ_λ curve (the AN's own scan, its Fig. 131, shows the interference structure near $\kappa_\lambda \approx$ 5–6); (ii) each minimisation is over nuisances only, hence robust; (iii) **the model may be rebuilt per point** — re-mixed templates *and* that point's theory band — because the minimiser never needs $\partial(\text{model})/\partial\kappa_\lambda$: the κ_λ-dependence of the band never enters any derivative, pull, or profiling step. It is bookkeeping *between* fits, not structure *within* one. **NSBI scans are pointwise in exactly the same way** (the prelim's profile-NLL scan already evaluates $t(\kappa_\lambda)$ on a grid over cached ratios), so the identical trick ports verbatim. Mild costs, engineering not conceptual: grid resolution (interpolate near thresholds and the σ-minimum) and per-point convergence.

C4 reassessed (12 Jul 2026): it splits — the rate part is closed, the shape part is an upgrade

C4a — the κ_λ-dependent *rate* band: closed (numerical, portable today). Because any scan (binned *or* NSBI) profiles at fixed κ_λ pointwise, a κ_λ-dependent lnN *magnitude* needs no POI-coupled per-event object: put band(κ_λ) (LHC HH WG / AN Table 40) into the green rate factor on $\nu_{\text{sig}}(\kappa_\lambda)$, profiled per scan point and correlated across points/runs exactly as CMS does. Since the theory uncertainty is the *leading* systematic and the rate band carries the −23%, **the dominant $b\bar{b}\tau\tau$ systematic is implementable in the NSBI fit at parity with CMS, today. C4b — the theory *shape* deformation: still open, but an *upgrade*, not a parity risk.** Scale/m_t variations deform each component's $d\sigma/dx$ (hence the discriminant) — the per-component $w_{\theta,s}(x;\alpha_\theta)$ morphing that *nobody* performs: CMS is normalisation-only. So C4b is not a gap NSBI must close to match the baseline; it is where *both* approaches currently truncate, and where NSBI's per-component structure is the natural upgrade. Table 40's **sign flip** ($+4/-18\%$ at κ_λ=1 $\rightarrow$ $+13/-4\%$ at κ_λ=5) is direct empirical evidence for the per-component picture: the net band rotates because the components' bands differ — exactly what blending POI-independent component bands with the known κ_λ coefficients reproduces automatically. **Gate refocus:** the C4 pilot is no longer "can we handle THU at all" but (i) source the component-differential bands, and (ii) measure the *truncation bias* — fit Asimov data generated with scale-varied *shapes* using the normalisation-only model and measure the κ_λ bias: small $\Rightarrow$ the CMS truncation is justified and C4b is deprioritised; large $\Rightarrow$ per-component shape morphing becomes a genuine physics improvement NSBI can claim (folded into studies #5/#19).

Best-case CI *width* vs. the binned fit (a complementary axis). The matrix above asks *can we do it*; this asks *what it buys on the interval width if we do* — assuming the morphing is calibrated and coverage correct (i.e. setting aside item 6's correctness question). Two baselines must be kept apart. *Relative to the stat-only NSBI number* (the preliminary $2.75\sigma \rightarrow 7.1\sigma$), honestly including systematics widens the interval for *both* NSBI and the binned fit — the stat-only figure was an underestimate. *Relative to the binned fit with the same NPs*, however, there is a clean result: binning a (multivariate) discriminant is a deterministic *coarsening* of the per-event likelihood, so by the data-processing inequality it cannot carry *more* information, and profiling nuisances preserves the ordering (the profiled POI information is a Schur complement, which is operator-monotone). Hence **best-case NSBI CI $\leq$ binned CI, systematics included** — properly handling a systematic can only *shrink NSBI's advantage toward parity*, never reverse it (a reversal requires a modelling/coverage failure, excluded here).

Mechanism of the shrinkage. Profiling a systematic costs POI information in proportion to how much the systematic's effect *aligns with the POI's* in the space actually used. NSBI's gain lives in the feature dimensions beyond m_{HH}; a systematic that also lives there *and* points like κ_λ gives part of that gain back, while one confined to m_{HH} (already captured by binning) or orthogonal to the extra features does not.

Challenge	**Binned (best case)**	**NSBI (best case)**	**Net effect on width advantage**
1 κ_λ–syst. m_{HH} degeneracy	profiling JES/τES eats κ_λ info (acute in m_{HH}; a binned-DNN recovers some)	extra features *break* the degeneracy ($\sim$12.8$\times$ →1.3$\times$ inflation)	**largest win**
2 $\sim$40 experimental NPs	each NP profiled per bin; width grows with count	same NPs, per-event; grows similarly	preserved (equal cost)
3 data-driven bkg systematics	per-bin nuisance on the template	full per-event S/B *if* high-dim ABCDnn + per-event morphing	preserved best-case; *eroded* if the DD bkg must be binned
4 κ_λ-coupled theory	κ_λ-dependent lnN/shape; widens at the σ-minimum	shared — a signal-*prediction* band, not reduced by per-event power	$\approx$ neutral (shared floor)
5 factorisation cross-term	re-run the DNN with both NPs shifted	non-factorised (quadratic) morphing	preserved (both handle it)
6 MC-stat of learned ratios	Barlow–Beeston template MC-stat	estimation variance of the *learned* morphing (finite $\pm1\sigma$)	**genuine give-back**
a shower via CARL/DCTR	conservative 2-point *envelope*	profiled, data-constrained learned NP	**win** (narrows the envelope)
b absorb b-tag/τ-ID SFs	external POG prior	in-situ standard-candle constraint	**win** (in-situ > prior)

Two caveats. (i) **Item 6 is the one genuine width give-back** even with perfect coverage: the morphing is a *learned* object estimated from the (smallest) $\pm1\sigma$ samples, so it carries a real estimation variance that a re-run-the-DNN template partly avoids — sample-starved $\pm1\sigma$ sets are where NSBI can pay *more* width (mitigate with more variation MC or an information-pooling GP morphing). (ii) The *realistic* binned baseline is a binned *DNN score*, not 1-D m_{HH}: the proven $\sim$2.5$\times$ gain was measured against binned m_{HH}, so NSBI's margin over a good binned-DNN is smaller, and the part that survives unconditionally is the pure *unbinning* increment (no information lost to binning the score).

Where the width gain survives

Best-case, NSBI is never *wider* than binned; the advantage is just non-uniform. **Biggest wins:** #1 (degeneracy broken by dimensionality) and the upsides (a,b) (conservative *envelopes/priors* become *in-likelihood* constraints). **Shared / neutral:** #4 (irreducible signal-theory cost), #2, #5 (same NP cost, both pay it). **The one give-back:** #6 (finite-sample MC-stat of the learned morphing). So the headline gain is most robust where it comes from a *real* information advantage (unbinning + degeneracy-breaking + in-fit constraint of over-conservative systematics), and it erodes only where the systematic is irreducible-to-both (#4) or must be *learned* from too-small samples (#6).

6 Statistics and inference

- **Data signal yield is tiny and intrinsic.** $\sigma \times \mathcal{B} \sim 2.5$ fb $\Rightarrow \mathcal{O}(10\text{–}100)$ selected signal events (Run-2 to HL-LHC), under large backgrounds — vs the off-shell tail at ~15% of σ. This is unchanged by MC and keeps the measurement systematics-comparable. (It is also the regime where NSBI's per-event information helps *most*.)
- **MC/training stats are ample** (≈500k–750k per κ_λ point), so the morphing/ratio NNs are well-trained — the "under-determined morphing" worry applies to the *systematic-variation* samples, not the nominal.
- **Validation & coverage.** CMS inherits decades-validated COMBINE (asymptotics + toys + Barlow–Beeston). NSBI needs a custom JAX/IMINUIT fit with an explicit Neyman construction and ratio-quality diagnostics (C2ST-like) built and validated from scratch [ATLAS p. 64,89].

7 Synthesis: could the NSBI κ_λ result be *worse* than the CMS fit?

Reading what CMS *actually* does removes a strawman: CMS does **not** bin m_{HH}; it bins a **multivariate DNN score** (shape systematics propagated by re-running the DNN on varied inputs, [CMS p. 93]). So CMS already carries the same multi-dimensional systematic information NSBI does — **NSBI is not "exposed to more systematics" than the real CMS baseline.** Against CMS-as-done, the comparison reduces to:

- **What NSBI buys:** *unbinning* (no information lost to binning the score) and a κ_λ-*optimal* per-event ratio (CMS's DNN is one fixed discriminant, not optimal at every κ_λ). These are the off-shell ~2× / FAIR-Universe ~20% gains.
- **What NSBI costs:** a *learned* morphing layer (NSBI must *train* the per-event systematic weight; CMS *counts* its re-run-DNN template, exact up to MC stat) and the *custom coverage validation*. The danger of "worse than CMS" is the *learned layer being mis-calibrated / under-validated* — producing a tighter-but-mis-covering interval — not a heavier systematics burden.
- **The genuinely new $b\bar{b}\tau\tau$ problem is the data-driven backgrounds** (Sec. 4): if QCD/-fakes cannot be given a faithful per-event density, the unbinned NSBI may have to bin them anyway, partially eroding its advantage in exactly the most sensitive channel.

Protections (so the worst case is "≈ CMS," not "worse and unknown"): honest morphing uncertainty (GP-style); explicit coverage validation; and **binning the NSBI discriminant as a built-in cross-check** (the binned fit is a coarsening of NSBI, so a regression is detectable and you fall back).

8 Summary of new challenges

	ATLAS off-shell 4ℓ	**κ_λ in HH→ $b\bar{b}\tau\tau$ (new challenge)**
POI	$\mu_{\text{off-shell}}$ (1 POI)	κ_λ ($+k_{2V}, k_V$, EFT); shape in m_{HH}
Final state	4 leptons, fully reco, clean	b-jets+τ_h+MET; SVfit; 3 channels
Backgrounds	one MC $q\bar{q}ZZ$ (interfering), CR-constrained	QCD (ABCD), DY (18-CR SFs), $t\bar{t}$ (CR-SF), **jet→ τ fakes**, single-H, diboson
Data-driven bkg in NSBI	none needed	**QCD & fakes have no per-event density**
Experimental systematics	few, small (leptons)	many, large (JES×11, τES, b-tag×5, DeepTau, ...)
Syst. vs signal shape	largely separable	**JES/τES deform m_{HH} (degenerate with κ_λ)**
Theory–POI coupling	none	**κ_λ-dependent theory band (Run-3: $+6/-23\%$ at SM; sign-flips with κ_λ)**
Factorisation of NPs	safe	JES×τES cross-term in m_{HH}
Signal yield	∼15% of σ	$\mathcal{O}$(10–100) events (fb-scale)
MC training stats	—	ample (≈500–750k/κ_λ); watch $\pm1\sigma$ samples
Stats framework	custom (built)	must port the custom fit; CMS has COMBINE

Bottom line

The NSBI *method* ports cleanly from off-shell to κ_λ. The new challenges are in the *inputs*, and rank roughly: (1) **data-driven backgrounds** (QCD, jet→ τ fakes) with no per-event density — the most awkward fit with NSBI, but *solved-with-caveats* by flow-based estimation (**ABCDnn**, Sec. 4); (2) the **κ_λ–systematic shape degeneracy** (JES/τES on m_{HH}) plus a much larger experimental-systematics surface, which makes *morphing accuracy + honesty* the limiting factor; (3) the **κ_λ-dependent theory** and **factorisation cross-terms** (non-factorisable morphing); (4) **irreducible single-Higgs**. None are NSBI-specific *except* the data-driven-background mismatch and the learned-morphing layer; the rest are equally CMS's problems, handled with the same NPs. Two scope caveats: NSBI's gain is in the *κ_λ/k_{2V} parameter constraint* (shape directions), *not* the SM-HH observation significance (rate-saturated, luminosity-set); and the genuine per-event edge is on *shape/correlated-operator* directions, not the rate-broken $\kappa_t \leftrightarrow \kappa_\lambda$ / $k_V \leftrightarrow k_{2V}$ ones.

Glossary of abbreviations & jargon

Every acronym and piece of jargon used in this note, in plain language. Symbols/couplings first, then abbreviations alphabetically.

Symbols & couplings

κ_λ (κ_λ) — Higgs trilinear self-coupling strength relative to the SM (= 1 in the SM); the parameter of interest. Enters gg→HH through the *triangle* diagram and interferes with the *box*.

k_{2V} — strength of the quartic $VVHH$ contact coupling (= 1 in SM); accessed in the high-m_{HH} VBF tail.

k_V — single Higgs–vector-boson coupling strength (VVH, = 1 in SM).

κ_t — top-quark Yukawa coupling strength (= 1 in SM); enters HH via the box and triangle.

m_{HH} — invariant mass of the di-Higgs system; the observable carrying the κ_λ interference shape.

$\cos\theta^*$ — production angle of the Higgs in the HH rest frame; the second variable in CMS's 2D κ_λ reweighting.

H_T — scalar sum of jet transverse momenta.

M_{jj} — invariant mass of the two VBF "tagging" jets.

N_j, N_b — number of jets / number of b-tagged jets (the ABCD(nn) conditioning variables).

p_T — transverse momentum (momentum in the plane transverse to the beam).

P_ref — reference hypothesis: the single fixed model all NSBI classifiers estimate density ratios *relative to*.

$r(x)$ — per-event likelihood (density) ratio — the quantity NSBI learns and the fit consumes.

ρ — (Pearson/Spearman) correlation coefficient; here between κ_λ and a systematic.

σ — standard deviation / Gaussian width ("$\pm 1\sigma$" = a one-standard-deviation systematic variation); also denotes a cross-section.

triangle / box — the two gg→HH amplitudes: the κ_λ-sensitive *triangle* (via an s-channel Higgs) and the κ_λ-independent *box*; their interference shapes m_{HH}.

GeV, TeV — giga-/tera-electronvolt (energy/mass units).

Abbreviations & jargon

ABCD — data-driven background estimate from four regions defined by two (assumed-independent) variables; the signal region $D = (B \times C)/A$.

ABCDnn — a normalizing-flow generalization of ABCD that morphs simulated background into the data-driven background *shape*, conditioned on extra variables.

AN — (CMS) Analysis Note: internal documentation (e.g. AN-18-121).

anchors — the fixed reference points (a κ_λ grid, or the $\pm 1\sigma$ systematic variations) between which a smooth morphing is interpolated.

Asimov dataset — a single artificial "data = expectation" dataset used to compute *expected* sensitivity without generating toys.

ATLAS, CMS — the two general-purpose LHC experiments.

AUC — area under the ROC curve; a classifier separation metric (0.5 = no separation).

Barlow–Beeston — standard treatment of finite-MC-statistics uncertainty via per-bin nuisance parameters.

BDT — boosted decision tree.

binned / unbinned — a fit that histograms an observable into template bins vs. one that uses each event's per-event likelihood directly (no binning). The CMS baseline is binned; NSBI is unbinned.

bootstrap — resampling with replacement to estimate a statistical uncertainty (e.g. of a trained network).

CARL / DCTR — the classifier-reweighting family: train a classifier to learn a per-event reweighting between two models. CARL = Calibrated Approximate Ratio of Likelihoods; DCTR = Deep Classifier-based ReweighTing.

closure (test) — a check that a method reproduces a known truth (e.g. predicted = true background in a validation region).

C2ST — Classifier Two-Sample Test: train a classifier to distinguish two samples; AUC ≈ 0.5 means they are indistinguishable (good closure).

Combine — CMS's standard binned maximum-likelihood fitting framework (HistFactory-style; asymptotic formulae plus toys, with Barlow–Beeston MC-stat handling).

CR / SR / VR — Control / Signal / Validation Region.

decay mode — the specific hadronic τ decay channel (e.g. 1-prong, 3-prong, with/without π^0); τES has per-decay-mode components.

DeepTau — CMS deep-learning hadronic-τ identification discriminant.

Delphes — a fast, parametrised detector simulation used for quick studies before full simulation.

DNN — deep neural network.

DY — Drell–Yan ($Z/\gamma^* \to \ell\ell$) background.

EFT / SMEFT — (Standard-Model) Effective Field Theory: parametrize new physics as higher-dimensional operators.

ES — energy scale (as in JES, τES).

EW — electroweak.

ensemble (spread) — training several networks (different seeds/folds); their spread estimates the MC-statistical uncertainty, propagated as one nuisance.

ExtendedABCD — ABCD augmented with extra sidebands to correct residual correlation between the two defining variables (a more accurate yield).

extrapolation (CR→SR) — applying a transfer/calibration learned in a control region to the signal region; its validity is the key data-driven-background caveat.

fake-factor — data-driven estimate of jet→ τ_h misidentification: a transfer factor measured in a τ-anti-isolated CR and applied to the SR.

Fisher information — quantifies how sharply an observable constrains a parameter; here used to show that added dimensions shrink the κ_λ–systematic degeneracy.

flavour-tagging — identifying a jet's originating quark (b/c/light) with an ML discriminant (b-tagging is the b-jet case).

F-score — harmonic mean of precision and recall; here a separation proxy.

ggF / VBF — gluon–gluon fusion / vector-boson fusion (the two HH production modes).

ggH, VBF H, VH, $t\bar{t}H$ (single Higgs) — the single-Higgs production modes (gluon fusion / vector-boson fusion / associated with a vector boson / with a top pair); the irreducible same-final-state background (distinct from ggF/VBF for HH).

GP — Gaussian Process: a non-parametric model giving predictions with calibrated uncertainty bands.

HH — di-Higgs (Higgs-pair) production; **HHVV** = the quartic $VVHH$ contact vertex (set by k_{2V}).

Hesse / Hessian — matrix of second derivatives of the NLL; its inverse gives parameter covariances (MINUIT's HESSE).

HL-LHC — High-Luminosity LHC.

ID — identification (e.g. τ-ID, b-ID).

iminuit — Python interface to the MINUIT minimizer used for the profile fit.

interference — the quantum cross-term between two amplitudes (here triangle×box in HH); it carries the κ_λ shape information.

ISR / FSR — initial-/final-state radiation (a parton-shower systematic).

JAX — an autodiff / accelerated-array library used to build the differentiable likelihood.

JER / JES — jet energy resolution / jet energy scale (systematics).

KL divergence — Kullback–Leibler divergence: a measure of difference between two probability distributions.

KS test — Kolmogorov–Smirnov goodness-of-fit test (a p-value for "same distribution").

LHC — Large Hadron Collider.

lnN — log-normal: the usual constraint shape for a *normalization* nuisance parameter.

LO / NLO — leading / next-to-leading order in perturbative QCD.

MAE — mean absolute error.

MC — Monte Carlo (simulated events).

ME — matrix element.

MET — missing transverse energy (momentum imbalance, here mainly from neutrinos).

ML — machine learning.

mixture model — writing the total per-event density as a weighted sum of signal/interference/background components (the NSBI fit model).

MMD — Maximum Mean Discrepancy: a distribution-matching loss (used to train ABCDnn).

morphing — smoothly deforming a per-event density or weight as a function of a parameter (κ_λ or a nuisance), built by interpolating between anchors.

NAF — Neural Autoregressive Flow: a flexible normalizing-flow density model (ABCDnn's network).

Neyman construction — the frequentist procedure for building confidence intervals with correct coverage.

NLL — negative log-likelihood.

NN — neural network.

NP — nuisance parameter: a non-POI (systematic) parameter that is profiled in the fit.

NSBI / SBI — (Neural) Simulation-Based Inference: inference from learned per-event likelihood ratios instead of binned templates.

PAS — (CMS) Physics Analysis Summary, a short public document.

PDF — parton distribution function (separately, also "probability density function").

Perlmutter — the NERSC GPU supercomputer used for training.

pivoting / decorrelation — training a classifier to be *insensitive* to a nuisance (statistically independent of it).

POG — (CMS) Physics Object Group; provides standard object calibrations / scale factors.

POI — parameter of interest (here κ_λ).

poly-exp — polynomial×exponential interpolation used between systematic ($\pm 1\sigma$) anchors.

Powheg — an NLO matrix-element event generator.

profile (likelihood) — maximizing the likelihood over the nuisance parameters at each value of the POI.

PS / PU — parton shower / pile-up (additional simultaneous pp collisions).

Pythia, Herwig — two parton-shower/hadronization generators; their difference gauges shower-modeling systematics.

QCD — quantum chromodynamics; "QCD multijet" = the data-driven multijet background.

RealNVP — a specific (identity-initializable) normalizing-flow architecture.

resolution — the smearing/precision of a measured quantity (cf. JER for jets); "resolution handles" = input features encoding it.

RMS — root-mean-square.

RS3L — Re-Simulation-based Self-Supervised Learning (a representation-robustness method).

Run-2 / Run-3 — LHC data-taking periods (Run-2: 2016–18; Run-3: 2022–25).

SF — scale factor: a data/MC correction applied to reconstructed objects (b-tag, τ-ID, . . .).

sideband — a region adjacent to (but outside) the signal region, used to estimate backgrounds; the extra regions in ExtendedABCD are sidebands.

SM — Standard Model.

Sophon — the (Sophon-AK4) jet "foundation model" backbone fine-tuned in the ML-pipeline studies.

SVfit — algorithm that reconstructs the $\tau\tau$ invariant mass from the visible decay products + MET.

THU$_{HH}$ — theory uncertainty on the HH cross-section/shape; κ_λ-dependent (Run-3 ggF scale+m_t: $+6/-23\%$ at the SM; the m_t-scheme asymmetry sign-flips across the κ_λ scan — AN-25-103 Table 40).

τ**ES (tauES)** — τ energy scale (a systematic).

toy / toy MC — synthetic pseudo-datasets drawn from the model, used for coverage and sensitivity tests.

trigger — the online selection deciding which collision events to record; carries its own efficiency scale factor.

VH / VZ — associated production of a Higgs / Z with a vector boson.

VIF — variance inflation factor: how much a parameter's variance grows due to correlation with others (a degeneracy metric).

WG — working group (e.g. the LHC HH WG).

ZZ, 4ℓ — a Z-boson pair / four charged leptons (the ATLAS off-shell final state).

References

[1] ATLAS Collaboration (J. Sandesara *et al.*), *Measurement of the Off-Shell Higgs Boson in $H \to ZZ \to 4\ell$ using Simulation-Based Inference*, ATLAS note HIGG-2023-01 (2024) = arXiv:2412.01600, Rep. Prog. Phys. 88 (2025). Page refs herein are to the note. Key: mixture model & systematics Sec. 5 (pp. 80–90); per-event factorised weights (pp. 83–84); MC-stat NP (pp. 86–87); JAX fit & impact (pp. 89–90).

[2] CMS Collaboration (C. Amendola *et al.*), *Search for Higgs boson pairs produced through gluon and vector boson fusion in the $b\bar{b}\tau\tau$ final state with Run-II data*, CMS AN-18-121 (2022). Key: signal reweighting Sec. 3.1 (p. 6); background estimation Sec. 7 (pp. 61–85; QCD ABCD pp. 61–62, DY 18 CRs pp. 63–78, $t\bar{t}$ CR-SF pp. 79–82); systematics Sec. 9 (pp. 90–97).

[3] S. Choi, J. Lim, H. Oh, *Data-driven Estimation of Background Distribution through Neural Autoregressive Flows (ABCDnn)*, arXiv:2008.03636 (2020). Production application: CMS all-hadronic four-top search, Run-2+3 (ExtendedABCD yield + ABCDnn shape for $t\bar{t}$+jets/QCD; obs. $3.95\,\sigma$, $\mu = 2.7^{+1.1}_{-0.9}$) — S. An (advisors M. Paulini, V. Dutta), Carnegie Mellon PhD thesis (2025), CMU KiltHub #29950337; MMD-trained NAF, conditioned on (N_{jets}, N_b), modelling the BDT+H_T shape. No public CMS-PAS located.

[4] A. Bahl, T. Plehn, N. Schmal, *global SMEFT fit of four di-boson processes with per-event likelihoods*, arXiv:2509.05409 (2025) — the ∼2× per-event gain on correlated-operator directions.

[5] ATLAS+CMS HH combination, arXiv:2602.23991 / CMS-PAS-HIG-25-014 (early 2026): $k_{2V} \in [0.73, 1.3]$, $\kappa_\lambda \in [-0.71, 6.1]$, $\mu_{HH} < 2.5$ (95% CL).

[6] A. Ghosh, M. Klute, J. Pan, *Focused Neural Simulation-Based Inference for the Higgs Self-Coupling in the $b\bar{b}\tau\tau$ Channel*, working note (22 Jun 2026); code: `NSBI-pheno/dihiggs_bbtautau`. Signal $gg \to HH$ from POWHEG at NLO with $\kappa_\lambda \in \{0, 1, 5\}$ (three-point quadratic morphing basis); reference $\kappa_\lambda = 5$; ten event-level observables; full background model with weighted fake-τ; unbinned extended likelihood vs binned m_{HH}, plus a focused test statistic. Detector response: **CMS full simulation** for the current validation, with a DELPHES fast-sim reproduction planned for the published phenomenological version (the setup the note describes). Results: $\kappa_\lambda = 0$ rejection 2.75σ ($450\,\text{fb}^{-1}$) → 7.1σ (HL-LHC); 68% interval ∼10× tighter than the binned fit; 500-toy coverage validated. *No experimental-systematic nuisances in the fit.*

[7] R. Schöfbeck *et al.*, *Unbinned measurements with machine-learned systematic uncertainties* (GOLLUM): method MLST 6 015007 (2025), tested PRD 112 052006 (2025). Per-event parametric NP morphing to arbitrary order, factorizable *and* NP×NP non-factorizable; tested on FAIR Universe (∼3 calibration NPs, simulated single-lepton data). Treats NPs as POI-*independent*.

[8] P. Shyamsundar, *Comment on "An implementation of neural simulation-based inference for parameter estimation in ATLAS"*, arXiv:2505.19156 (2025). Argues, via a toy counterexample, that the ATLAS double-bootstrap estimator of the finite-simulation (MC-stat) uncertainty does not capture it — a correctness critique, not merely an under-validation. Bears on item 6.

[9] S. Benevedes, J. Thaler, *Frequentist Uncertainties on Neural Density Ratios with wifi Ensembles*, arXiv:2506.00113 (2025). Trains basis functions $f_i(x)$ and propagates the weight covariance *analytically* (no bootstrap) to an asymptotic frequentist uncertainty band on a learned density ratio; empirically correct coverage on a q/g SBI example. Leading candidate estimator for item 6 — addresses both the correctness critique *and* the ensemble/bootstrap compute multiplier; per-event band also feeds the honest morphing-uncertainty of item 9/#15. Untested on interference/morphing/weighted-MC.

[10] *Neural Fake Factor Estimation* (data-based density-ratio inference), arXiv:2511.06972, JHEP 04 (2026) 188. Per-event continuous fake factor as a learned density ratio (NSBI-compatible); *defers* fake-rate systematics to future work. Bears on items 3 and the data-driven backgrounds of Sec. 4.

[11] CMS Collaboration (A. Andrade *et al.*), *Search for non-resonant Higgs boson pair production in the $b\bar{b}\tau\tau$ final state with Run-3 data*, CMS AN-25-103 (draft, 2026); $172.3\,\mathrm{fb}^{-1}$ at 13.6 TeV + Run-2 combination. Key for this note: exact 3-component ggF signal morphing at fit time (Eq. 52–55; NLO samples at $\kappa_\lambda = 1, 2.45, 5$; VBF 6-component analog); theory uncertainties as *normalisation-only* lnN (Sec. 7.1, Table 42; ggF scale+$m_t = +6/-23\%$ at SM — the leading $b\bar{b}\tau\tau$ systematic); the band tabulated *per κ_λ point* with the m_t asymmetry sign-flipping (Table 40); pointwise scans and limit-vs-σ_{theory} exclusions (Secs. 8.4.5–8.4.6); joint QCD+m_t correlated across runs (Sec. 8.5.1). Basis of the C4a/C4b split in Sec. 5.